Placing Defects Where You Ask: Location-Specific Defect Generation in a Model Glass with Conditional Diffusion

Stacy Cheruto Nicholaus Ladislaus Abubakar Chilala
Gentille Uwera Michel Sangwa Boniface Godwin
College of Engineering, Carnegie Mellon University Africa
{ccheruto, eladisla, achilala, guwera, sbmichel, bgodwin}@andrew.cmu.edu

Abstract

Defects control many properties of amorphous materials, but obtaining a specific defect at a chosen location still means waiting for it to form in a slow melt-quench simulation or inserting it by hand. Diffusion models now generate realistic amorphous structures, yet they cannot be asked for a particular defect at a particular place. In a two-dimensional Kob–Andersen glass, a periodic E(n)-equivariant denoiser receives pinned atoms and a per-atom defect label, uses classifier-free guidance, and regenerates a disk around a requested site inside an existing glass. Every sample is relaxed before it is scored, and its surroundings are compared with those of naturally formed defects. Averaged over three training runs, the model places under-coordinated defects 3.5× as often as hand insertion (0.22 vs. 0.06) with near-natural surroundings, and B-rich defects 1.5–2.4× as often (0.45–0.71 vs. 0.30) with surroundings as natural or more natural. Local melt-quench of the same disk almost never succeeds. Over-coordinated defects fail for every method, and we show that nothing local can hold them. The ranking survives every evaluation setting we varied. Along the way we find that the sampler's step count acts as a cooling rate, that training on the full noise range the sampler visits decides whether generated defects survive relaxation, and that training runs with the same validation loss differ in their effective guidance strength.

1 Introduction

A defect's behaviour in a glass depends on its surroundings, so studies of plasticity, relaxation or optical properties need many realistic examples of a given defect. Today these come from two sources: melt-quench molecular dynamics (MD), where one waits for the defect to form by chance at an uncontrolled place, and hand insertion, where atoms are moved or replaced and the structure relaxed, which can leave artificial surroundings. Generative models offer a third route. DM2 [1] denoises random atomic positions into amorphous silica and metallic glasses whose structure and properties match MD, at a fraction of the cost for slow cooling rates. Like most generative models for materials [2–6], it conditions on global quantities. It cannot be asked for a particular local feature at a particular place.

We frame location-specific defect generation like image inpainting [7]: the defect is fixed, and the model generates realistic surroundings. Two points shape the study.

- *Location must mean something.* "At a chosen location" is trivial in an empty periodic box, since any naturally formed defect can be translated to any point. Our headline task therefore regenerates a disk around a requested site *inside an existing glass*, with everything outside the disk held fixed (Task B). This is the setting of host-guided inpainting for crystals [8] and of linker design between fixed fragments [9].
- *Success must survive physics.* A generated defect counts only if it is still there after the whole structure has been relaxed to a local energy minimum. Realism is measured against naturally formed defects of the same class, with a natural-vs.-natural floor as the reference.

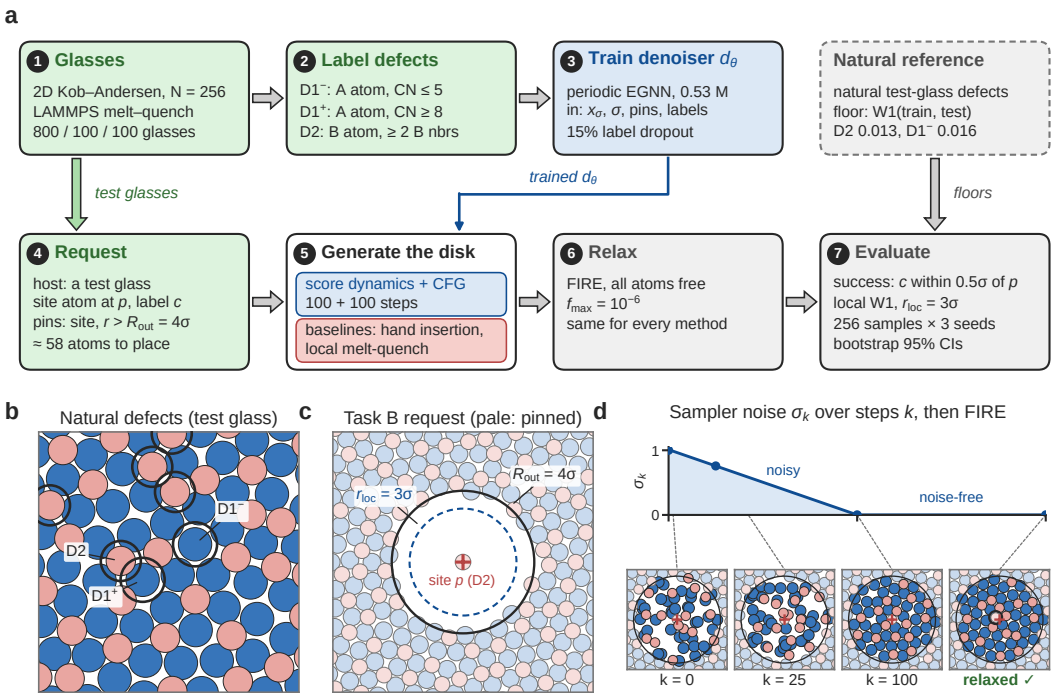


Figure 1: Study overview. **(a)** Workflow: labelled melt-quench glasses (1–2) train a periodic EGNN denoiser conditioned on pins and per-atom labels (3); a request (4) is answered by guided score dynamics or a baseline (5), relaxed with all atoms free (6) and scored against natural defects (7). **(b)** Natural defects (rings) in a test glass; A atoms blue, B atoms pink and smaller. **(c)** A real Task B request: pinned atoms (pale, including the site atom under the cross), the regenerated disk (R_{out}) and the realism window (r_{loc}). **(d)** The model ($w = 6$) answering it: free atoms start uniform in the disk and are denoised as σ_k falls; after relaxation the requested D2 (ring) is at the site.

Contributions.

1. A task and evaluation protocol for location-specific defect generation in glasses: host-conditioned inpainting, success after relaxation, local realism against natural defects, and an MD baseline that re-melts the same disk.
2. A conditional periodic EGNN with pinned atoms, per-atom defect labels and classifier-free guidance [10]. It beats hand insertion and local melt-quench for two of three defect classes, and the result holds across evaluation settings, re-sampling and three training seeds.
3. Lessons for diffusion models of glasses: the number of sampler steps acts as a cooling rate; the training noise range must cover the sampler's starting noise; training runs differ in effective guidance strength, which the denoising loss does not show; equivariance matters more than validation loss for sample quality; and over-coordinated defects are held by their host beyond 3.5σ, so no local method can place them.

Figure 1 summarises the study.

2 Related work

Diffusion models for materials. Denoising diffusion and score-based models [11, 12] now generate crystals [2–4] and disordered solids. For amorphous materials, DM2 [1] learns to remove Gaussian displacements from glass configurations and samples by a noisy relaxation it calls score dynamics. It conditions on the cooling rate and reproduces structure and mechanical properties of silica and metallic glasses. Other work conditions on spectra [6] or target properties [5]. All of these control global quantities; none places a chosen local motif at a chosen site.

Local control. In images, RePaint [7] inpaints a masked region with an unconditional model, and guidance [10, 13] strengthens a condition at sampling time. In materials and molecules, the closest settings are host-guided inpainting for crystal structure prediction [8], constrained diffusion that relaxes crystals containing given point defects [14], and linker generation between fixed molecular fragments [9]. A glass differs from these in two ways. There is no lattice to define a defect against, so a defect is a local motif that also occurs naturally. And realism can only be judged statistically, against the ensemble of naturally formed motifs. Our evaluation is built around both points.

3 Setup

Glass and defects. We use the two-dimensional Kob–Andersen binary Lennard-Jones mixture [15, 16]: 65:35 A:B at density 1.2, with $N = 256$ atoms in a periodic square box of side 14.6σ. Glasses are made in LAMMPS [17] by melting at $T = 2.0$ for 100τ, quenching at 10^{-2} per τ to $T = 0.01$ and minimising the energy: 800 training, 100 validation and 100 test glasses, each screened for crystallinity. Defects are defined by cutoff coordination, with the first shell within $1.4\,\sigma_{\alpha\beta}$:

- **D1**$^{-}$: an A atom with at most 5 neighbours (1.7 per glass);
- **D1**$^{+}$: an A atom with at least 8 neighbours (1.4 per glass);
- **D2**: a B atom with at least two B neighbours within 1.2σ (10 per glass).

Thresholds are frozen on the training split.

Tasks. In **Task B** (headline), a test glass supplies the host. The requested site p is the host atom of the defect's species nearest a random point. That atom, which carries the label, and every atom farther than $R_{\text{out}} = 4\sigma$ from it are pinned; the ≈58 atoms in between are regenerated, starting from uniform random positions. **Task A** generates a whole box with the defect at its centre. It is reported for completeness, since location is trivial there.

Denoiser architecture. The denoiser $d_\theta(x_\sigma, \sigma, m, c)$ (Figure 2a) takes a noisy configuration and returns one 2D vector per atom: its estimate of the displacement that the noise added. Rotating, translating or relabelling the atoms of a glass must rotate, leave unchanged or relabel these vectors in the same way. We build these symmetries into the network rather than learning them, using an E(n)-equivariant graph neural network (EGNN) [18] adapted to periodic boundaries. It has four parts.

- **Periodic graph.** Every atom is a node joined to all atoms within $r_c = 2.5\sigma$ (about 24 neighbours). Edge vectors r_{ij} follow the minimum-image convention, so atoms near opposite faces of the box are neighbours. Distances enter through 16 Gaussian radial basis functions (RBFs) and a cosine cutoff f_c, so a message switches on smoothly as two atoms approach.
- **Node embedding.** Learned embeddings of the species t_i, the pin flag m_i and the label token c_i ("none" everywhere except the requested site, where it names the defect class) are added to an MLP of random Fourier features of $\log\sigma$ and passed through an MLP, giving $h_i^0 \in \mathbb{R}^{128}$. The noise level tells the network how far atoms may sit from clean positions; the label tells it what to build at the site.
- **EGNN layers.** Four layers (Figure 2b) each compute messages, a residual feature update and an output vector:

$$
\begin{aligned}
m_{ij} &= \phi_e\big(h_i, h_j, \mathrm{rbf}(|r_{ij}|)\big)\, f_c(|r_{ij}|), \\
h_i &\leftarrow h_i + \phi_h\Big(\mathrm{LN}(h_i),\ \tfrac{1}{K_0}\textstyle\sum_j g(m_{ij})\, m_{ij}\Big), \\
v_i^{\ell} &= \tfrac{1}{K_0}\textstyle\sum_j r_{ij}\, \phi_x(m_{ij})\, f_c(|r_{ij}|),
\end{aligned}
\tag{1}
$$

  where ϕ_e, ϕ_h, ϕ_x are two-layer SiLU MLPs, g a sigmoid gate, LN layer normalisation and $K_0 = 24$. Features and messages depend only on distances and are invariant; the only vectors are the edge vectors, so v_i^{ℓ}, a weighted sum of them, rotates exactly with the glass. Unlike the original EGNN, positions are not moved inside the network: each layer adds its own vector to the output.

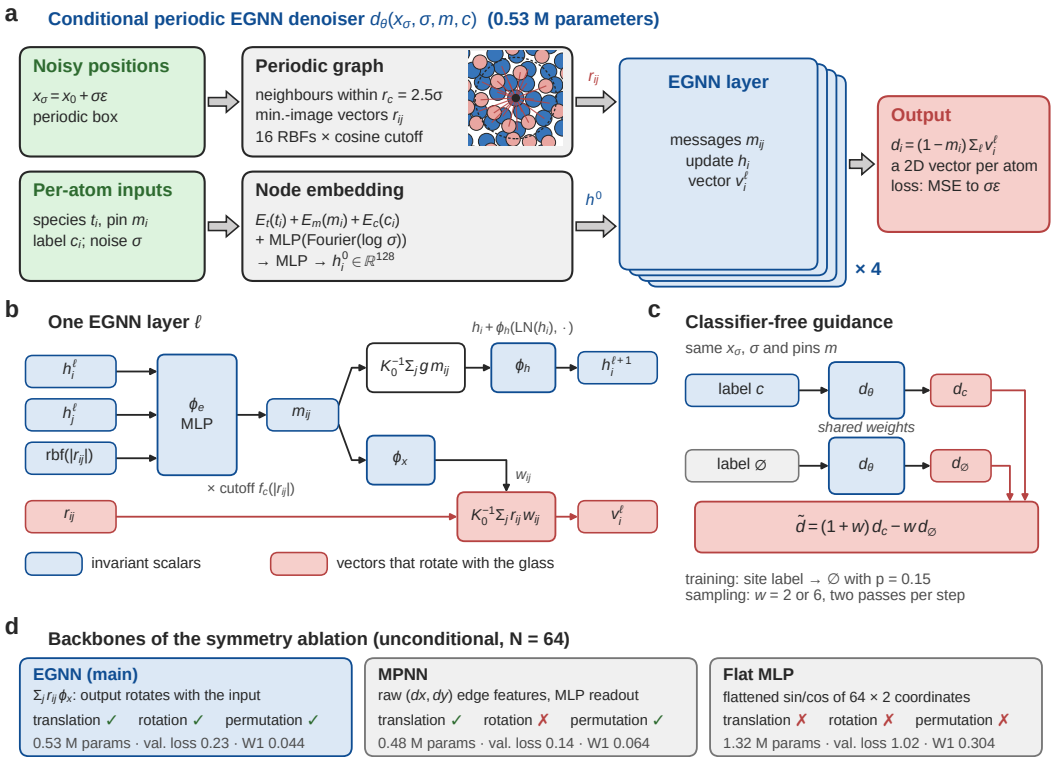


Figure 2: The deep learning models. **(a)** The conditional denoiser: noisy positions define a periodic graph, the other inputs define node features, and four EGNN layers turn both into one displacement vector per atom. **(b)** One EGNN layer (Eq. 1): blue tensors are invariant scalars, red tensors rotate with the glass; every output vector is a sum of edge vectors with invariant weights, so the prediction rotates exactly with the input. **(c)** Classifier-free guidance runs the same network with and without the site label. **(d)** The backbones of the symmetry ablation (Table 3).

- **Output.** $d_i = (1 - m_i)\sum_\ell v_i^\ell$, zero on pinned atoms. It is translation-invariant and equivariant to rotations, reflections and permutations by construction; `tests/test_equivariance.py` checks translations, rotations and permutations numerically.

The network has 0.53 M parameters, 59 k of them in the embedding. Because the condition enters only through node features, one network serves whole-box, patch and host requests. Classifier-free guidance (Figure 2c) needs no second network: the same weights are evaluated with and without the site label.

Ablation backbones. To test what the symmetries are worth (Section 4.4), we trained two less symmetric denoisers with the same inputs and recipe (Figure 2d). The *MPNN* [19] keeps the graph and message function but adds the raw components (dx, dy) of r_{ij} to the edge features and reads the output from h_i with an MLP, so it is permutation-equivariant but not rotation-equivariant (0.48 M parameters). The *flat MLP* concatenates, for all 64 atoms sorted by species, the sine and cosine of the fractional coordinates and one-hot species, pin and label, appends the σ embedding, and maps this through four 512-wide SiLU layers to 128 outputs (1.32 M parameters). It is periodic but has no other symmetry.

Training. We follow DM2's formulation [1]. Positions are perturbed as $x_\sigma = x_0 + \sigma\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I)$ (pinned atoms untouched), and the denoiser d_θ regresses the added displacement on the unpinned atoms U:

$$\mathcal{L}(\theta) = \mathbb{E}_{x_0,\sigma,\varepsilon,m,c}\left[\frac{1}{|U|}\sum_{i\in U}\left\| d_\theta(x_\sigma, \sigma, m, c)_i - \sigma\varepsilon_i \right\|^2\right], \tag{2}$$

where m is the pin mask and c the per-atom labels. Further details:

- $\sigma \sim \mathcal{U}(10^{-3}, \sigma_{\max})$;
- training requests mix whole-box, patch and host-inpainting modes (probabilities 0.5 / 0.3 / 0.2), centred on a natural defect half of the time;
- labels are dropped with probability 0.15 for classifier-free guidance;
- AdamW, learning rate 2×10^{-4}, batch 16, gradient clipping at 1, 50,000 updates, weights averaged with EMA 0.999; about 4 h on one T4 GPU.

Two recipes differ only in $\sigma_{\max}$: 0.5 (DM2's choice of half a bond length; used for D1$^-$ and Task A) and 1.0 (Section 4.2; used for D2).

Sampler. DM2's score dynamics alternates denoising with fresh noise,

$$x_{k+1} = x_k - \tilde{d}(x_k, \sigma_k) - \eta_k, \qquad \eta_k \sim \mathcal{N}(0, s_k^2 I), \quad s_k \sim \mathcal{U}(10^{-3}, \sigma_k), \tag{3}$$

with pinned atoms reset after every step. Classifier-free guidance [10] combines the conditional and label-free predictions,

$$\tilde{d} = (1+w)\, d_\theta(x, \sigma, m, c) - w\, d_\theta(x, \sigma, m, \varnothing), \tag{4}$$

so $w = 0$ is the unguided conditional model. We use 100 steps with σ_k falling linearly from 1.0 to 10^{-3}, then 100 noise-free steps; the σ passed to the model is clipped at its training $\sigma_{\max}$.

Baselines.

- *Chance*: inpainting the same disk without a label.
- *Hand insertion* (current practice): a natural defect patch of radius 1.5σ, drawn from the training split, is pasted at the site; each patch atom replaces the nearest host atom of its species.
- *Local melt-quench*: in LAMMPS, atoms beyond R_{out} are frozen, the disk is melted at $T = 2.0$ and quenched at the dataset rate, i.e. waiting for the defect, confined to the disk.
- For Task A: *Clamp* and *RePaint* [7] with an unconditional model.

Evaluation.

- **Relaxation.** Every sample is relaxed with FIRE [20] ($f_{\max} = 10^{-6}$), with all atoms free, host included.
- **Success.** The atom of the requested species nearest the site lies within $r_{\text{tol}} = 0.5\sigma$ of it and carries the requested label.
- **Local realism.** The Wasserstein-1 distance (W1) between the per-atom energies in a window of radius $r_{\text{loc}} = 3\sigma$ around the site and those around natural defects of the same class in the test glasses. The floor is natural training defects against natural test defects: D1$^-$ 0.016, D1$^+$ 0.018, D2 0.013.
- **Statistics.** 256 samples per row, with 95% bootstrap confidence intervals (CIs) over samples.

4 Results

4.1 Main result: defects inside existing glasses

Table 1 and Figure 4 give the Task B results. Model rows are means $\pm$ standard deviations over three training runs (seeds 0–2), each with 256 samples.

The model beats both baselines on the two classes that can be placed.

- **D2.** At $w = 6$ the model succeeds 2.4$\times$ as often as hand insertion on average (2.1–2.7$\times$ per seed), with surroundings as natural or more natural. At $w = 2$ it succeeds 1.5$\times$ as often (1.2–1.9$\times$) with clearly more natural surroundings for every seed (W1 0.045–0.056 vs. 0.087).
- **D1$^-$.** The model succeeds 3.5$\times$ as often as hand insertion at $w = 2$, with surroundings 1.2–1.5$\times$ the natural floor.

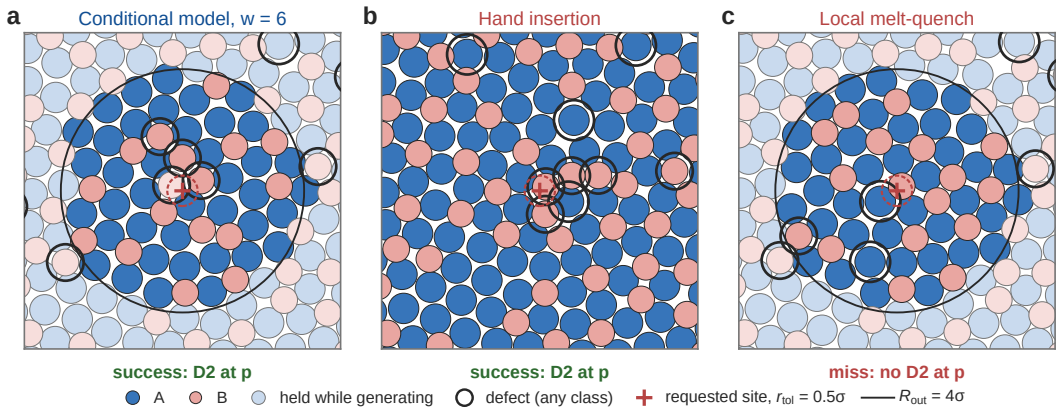


Figure 3: The request of Figure 1c answered by each method, after relaxation ($\pm 5\sigma$ around the site; pale: held fixed while generating; rings: defects). **(a)** The model places a D2 at the site. **(b)** Hand insertion also succeeds here, but its surroundings are far from natural on average. **(c)** Local melt-quench re-forms the disk without one. Of the six requests drawn for these figures, the model answered two, hand insertion four and local melt-quench one; Table 1 gives rates over 256 requests per seed.

Table 1: Task B, success after relaxation and local realism (W1 of per-atom energy in a 3σ window; lower is more natural). Model rows: mean $\pm$ s.d. over three training runs; baselines: estimate with 95% CI. D2 uses the $\sigma_{\max} = 1.0$ recipe, D1$^-$ and D1$^+$ the $\sigma_{\max} = 0.5$ recipe. †2,900-step sampler. ‡Seed 0 only.

	D2 (B-rich)		D1$^-$ (under-coordinated)		D1$^+$
Method	success	local W1	success	local W1	success
Chance (no label)†	0.03	0.062	0.02	0.053	0.00
Local melt-quench	0.05 [0.03, 0.08]	0.065	0.00 [0.00, 0.00]	0.024	0.01
Hand insertion	0.30 [0.24, 0.36]	0.087	0.06 [0.04, 0.09]	0.032	0.02
Model, $w = 2$	0.45 $\pm$ 0.11	**0.051** $\pm$ 0.006	**0.22** $\pm$ 0.02	**0.022** $\pm$ 0.002	0.03‡
Model, $w = 6$	**0.71** $\pm$ 0.08	0.072 $\pm$ 0.016	0.21 $\pm$ 0.04	0.038 $\pm$ 0.004	0.04‡
Natural floor	—	0.013	—	0.016	—

- **Local melt-quench** almost never produces the defect at the site. Re-forming a 60-atom disk at the dataset's cooling rate is the "wait for it" approach in miniature, and rare defects remain rare.
- **Cost.** Per success the model needs 1–4 GPU-seconds, comparable with hand insertion (0.8–3.6 s, almost all relaxation) and far below local melt-quench (65 s per D2 success; no D1$^-$ success at all).

Guidance trades realism for success. Stronger guidance places the defect more often before relaxation: D2 0.86 at $w = 2$ against 1.00 at $w = 6$ (seed 0). More of those placements also survive relaxation, but the surroundings move away from natural. For D1$^-$, success saturates at $w = 2$: extra guidance puts more defects in place before relaxation (0.85 at $w = 6$), but they sit in shallow basins and relax away. Without guidance ($w = 0$) success drops (D1$^-$ 0.09 vs. 0.23, D2 0.24 vs. 0.30 at 300 steps; Appendix A).

D1$^+$ fails for every method, and the reason is not local. Natural D1$^+$ sites survive noise and relaxation in their own glass, yet a natural D1$^+$ transplanted into a new host with its 47 nearest atoms (radius 3.5σ) relaxes away 97% of the time (Appendix B). Its host holds it beyond 3.5σ, so neither a patch nor a pinned label can carry it.

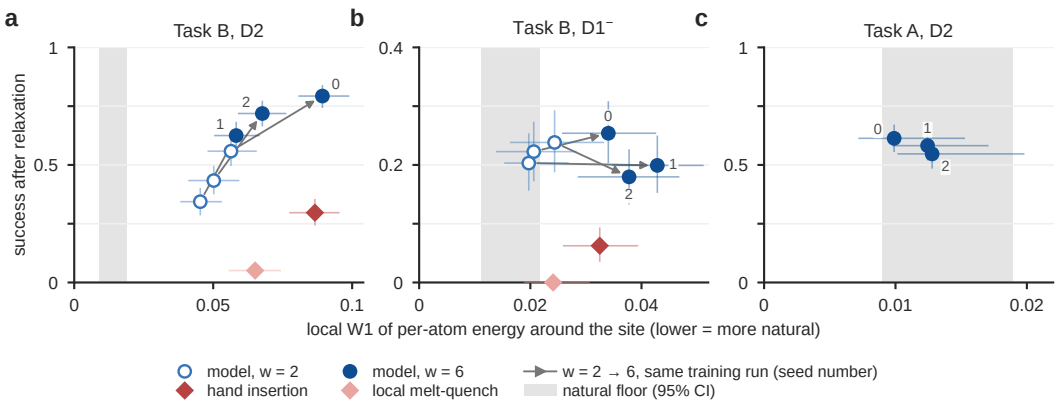


Figure 4: Success against local W1 for three training runs. Open circles: $w = 2$; filled: $w = 6$; arrows join the two weights of one run (seed number). Diamonds: baselines; grey band: natural floor; lines: 95% CIs; panel (b) has its own success scale. For D2, every run moves along the same success–realism line as guidance grows.

4.2 Why generated defects survive relaxation, or do not

A natural defect survives Gaussian noise of mean displacement up to $\approx 0.19\sigma$ followed by relaxation 99–100% of the time (Appendix B). Generated samples move 0.16–0.24σ during relaxation, yet only 25–80% of their placed defects survive. The gap is the generator's: its defects sit in shallower basins than natural ones.

The training noise range is the main lever (Table 4, Appendix A). Following DM2, our original recipe trains with noise up to half a bond length ($\sigma_{\max} = 0.5$), while the sampler starts at $\sigma = 1$, a range the model never saw. Retraining with $\sigma_{\max} = 1.0$ makes the model place the D2 defect more reliably before relaxation, and keep it more often afterwards: 79% instead of 57% at $w = 6$. D2 success rises from 0.54 to 0.79 (seed 0). For D1$^-$ the original recipe remains slightly better (0.22 vs. 0.21 at $w = 2$, 0.25 vs. 0.22 at $w = 6$), so we report each class with its better recipe.

4.3 Robustness

Reproducibility and training seeds. Re-running the nine headline rows with the same seed reproduced eight of them to the last digit; the ninth moved within its CI. A fresh evaluation draw (new hosts, sites and noise) moved success by 0.02 on average, within the bootstrap CIs. Across three training runs, D1$^-$ and Task A vary no more than the evaluation noise (s.d. 0.02–0.04), while Task B D2 varies twice as much (s.d. 0.08–0.11; Figure 4). The D2 seeds rank identically in success and in local W1, and their placement rate before relaxation differs (0.41–0.86 at $w = 2$), although their validation losses agree to within a few percent. Each training run thus ends with a different *effective guidance strength*, invisible to the denoising loss. The model beats both baselines for every seed, with one exception: seed 1 at $w = 2$, whose success CI overlaps hand insertion's while its surroundings remain clearly more natural.

Evaluation settings. We moved each setting on its own (Figure 5, Appendix A): the defect cutoffs by ± 0.05 (relabelling the natural references to match), the success radius by $\pm 0.1\sigma$, the realism window by $\pm 0.5\sigma$, and the regenerated disk R_{out} to 3 and 5σ. The ranking never changes, but absolute success moves by up to ≈ 0.1 with the definitions, so rates must be quoted with the definition used. Hand insertion's damage is local: its W1 falls from 0.113 to 0.068 as the window grows, while the model's is spread over the regenerated disk (W1 0.089 at $w = 6$ throughout), so realism should be compared at more than one window size. Success changes little with R_{out} (within the CIs), although the model was trained at 4σ only, and a larger disk makes D2 surroundings much more natural (W1 0.026 at $R_{\text{out}} = 5$, $w = 2$), which may partly reflect the boundary between regenerated and pinned atoms moving out of the window.

4.4 Lessons for diffusion models of glasses

The step count is a cooling rate. The score-dynamics sampler adds fresh noise at every step, so a longer schedule is a slower anneal (Table 2, Appendix A). DM2's default of 3,000 steps over-anneals our glass: lower energy than the training glasses, fewer B–B contacts, worse $g(r)$. 100 steps reproduce the natural energy at a fifteenth of the cost, without changing Task B success.

DM2's own pretrained silica model shows the same issue in the other direction: its samples have the right tetrahedra, but their disorder matches a glass cooled 100× faster than its training data (Appendix C). The cooling rate of generated glass should be measured, not assumed.

Whole generated glasses are chemically biased. Even at 100 steps, generated boxes have about two-thirds of the natural B–B contacts, whatever the noise range, which is why Task B, keeping most of a natural host, is the more trustworthy setting. In Task A the model still reaches D2 success 0.58 ± 0.03 with surroundings at the natural floor (W1 0.012 ± 0.002), against 0.34–0.35 for Clamp and RePaint patches.

Symmetry matters more than loss. Trained with the same recipe at $N = 64$ (Figure 2d; Table 3, Appendix A), the EGNN makes the most natural glasses, the MPNN reaches a *lower* validation loss but worse samples, and the flat MLP does not learn at all (validation loss at the predict-zero level; 16% of its samples partly crystalline).

5 Discussion and limitations

For two of three defect classes in a model glass, a diffusion model can be asked for a specific defect at a specific place, with natural-looking surroundings, more reliably than the methods used today. The failure is informative: an over-coordinated atom is stabilised by its host beyond any local patch, a limit for every local placement method, not just ours.

What this means in practice. When many instances of a local motif are needed in realistic surroundings, the conditional model gives 1.5–3.5× more correct placements per attempt than hand insertion, with surroundings closer to natural; hand insertion remains the cheapest per attempt and is adequate when the surroundings do not matter, and re-melting a region is not a practical way to obtain a specific defect. The guidance weight is a dial between success and realism (Figure 4); since each trained model sits at its own point on that curve, it should be chosen per model rather than fixed in advance. A defect stabilised by its host beyond 3.5σ, like $D1^+$, cannot be carried by a patch or a label on one atom: regenerating a larger region, which already improves realism, or conditioning on a descriptor of the wider host are the obvious next tests. Finally, scoring after relaxation, natural floors, and several seeds and settings apply to any generative model for disordered matter; each changed a conclusion here.

Limitations and outlook. The system is a 256-atom two-dimensional model glass, defects are defined geometrically, whole generated glasses are chemically biased, and D2 success varies between training runs (choosing w per model on validation data may help; untested). Because MD is cheap in 2D, cost is not a decisive argument here. We are extending the approach to three-dimensional amorphous silica with a NequIP-type denoiser [21] and DM2's data and potential [22], targeting coordination defects, non-bridging oxygens and small rings, which are rare in slowly cooled glass.

6 Conclusion

A periodic equivariant denoiser with pinned atoms, per-atom labels and guidance places requested defects inside existing glasses more reliably than hand insertion or local melt-quench, with near-natural surroundings. Sampler length, noise range and training seed all shape its output and must be measured, not assumed.

Code and data. Code, generated datasets, trained weights and every result in this paper: https://github.com/chilalaabubakar/IDL-PROJECT.

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A Additional results

Guidance sweep (Task B, seed 0, 300 steps, $\sigma_{\max} = 0.5$). The table gives success after relaxation, with placement before relaxation in brackets.

w	0	1	2	4	6
$D1^-$ success (placed)	0.09 (0.19)	0.17 (0.46)	0.23 (0.59)	0.21 (0.77)	0.23 (0.85)
D2 success (placed)	0.24 (0.32)	0.31 (0.58)	0.30 (0.73)	0.47 (0.80)	0.56 (0.97)

Task A (seed 0, 100 steps). Success after relaxation, with local W1 in brackets.

Method	D2	$D1^-$	$D1^+$
Unconditional (chance)	0.01 (0.055)	0.00 (0.069)	0.00
Clamp patch	0.34 (0.022)	0.01 (0.053)	0.04
RePaint patch ($U = 1$)	0.35 (0.045)	0.02 (0.042)	0.04
Model $w = 6$, $\sigma_{\max} = 0.5$	0.61 (0.010)	0.23 (0.034)	0.05
Model $w = 6$, $\sigma_{\max} = 1.0$	0.84 (0.046)	0.16 (0.092)	0.01

RePaint resampling ($U = 5, 10$) does not help on this sampler: every resampling pass re-runs part of the noisy schedule, so it acts as extra annealing (energy -3.704 and -3.712).

Table 2: Unconditional whole-box samples (seed 0) against sampler steps. Natural test glasses: energy -3.681, g_{AB} floor 0.067, 0.56 B–B contacts per B atom.

noisy steps	100	300	900	2,900 (DM2)
energy per atom	-3.678	-3.692	-3.706	-3.720
$g_{AB}(r)$ L1	0.23	0.30	0.39	0.49
B–B contacts per B	0.36	0.31	0.25	—

Table 3: Symmetry ladder at $N = 64$ (unconditional, 100 steps, relaxed). Natural: energy -3.636, g_{AB} floor 0.094, 0.475 B–B per B, local W1 floor 0.007.

Model	energy	g_{AB} L1	B–B per B	local W1	crystallised	val. loss / σ^2
EGNN	-3.626	**0.21**	**0.32**	**0.044**	0%	0.23
MPNN	-3.645	0.25	0.26	0.064	0%	0.14
Flat MLP	-3.341	0.67	1.31	0.304	16%	1.02

B Stability of natural defects

Noise followed by relaxation. All atoms of 100 test glasses were displaced by Gaussian noise, the glasses relaxed, and each natural defect checked with the success test. The table gives the fraction of defects kept.

noise σ (mean displacement)	0.05 (0.06)	0.10 (0.12)	0.15 (0.19)	0.20 (0.26)	0.30 (0.48)
$D1^-$ (155 sites)	1.00	1.00	0.99	0.68	0.08
$D1^+$ (135 sites)	1.00	0.99	0.99	0.69	0.06
D2 (1,010 sites)	1.00	0.99	1.00	0.85	0.29

Transplanting $D1^+$. Natural $D1^+$ patches were pasted into test hosts and relaxed, 64 insertions each. All are in place before relaxation; afterwards:

- radius 1.5σ (9 atoms): 0.00;
- radius 2.5σ (23 atoms): 0.05;
- radius 3.5σ (47 atoms): 0.03.

C Reproductions

DM2 on amorphous silica. We reran DM2's pretrained unconditional model with its demo settings: a 300-atom box, 2,900 noisy and 100 final steps, four seeds, on CPU. We compared the samples with its 1 K/ps training glasses.

- **Local structure is right.** Si is 99.3% 4-fold coordinated, the Si–O bond is 1.650 Å (training 1.646 Å), O–Si–O is 109.4°, and every $g(r)$ peak is in place.
- **Disorder is too high.** At matched sample size, the $g(r)$ distance to the training glasses (Si–O 0.58) exceeds the held-out floor (0.31) and matches a 100 K/ps quench (0.50). The O–Si–O width (6.4° vs. 5.6°) and edge-sharing tetrahedra (0.0125 vs. 0.0020 per Si) agree.

Course proof-of-concept. The provided notebook (an MLP denoiser on a jittered 3×3 grid) reproduces the figure in our proposal over five seeds:

- pair-distance W1 0.063 (range 0.033–0.122; proposal 0.068);
- generated disorder 0.153 (proposal 0.163).

Its antisite demo trains on overlapping atoms because of integer site coordinates; a one-line fix is documented in the repository.

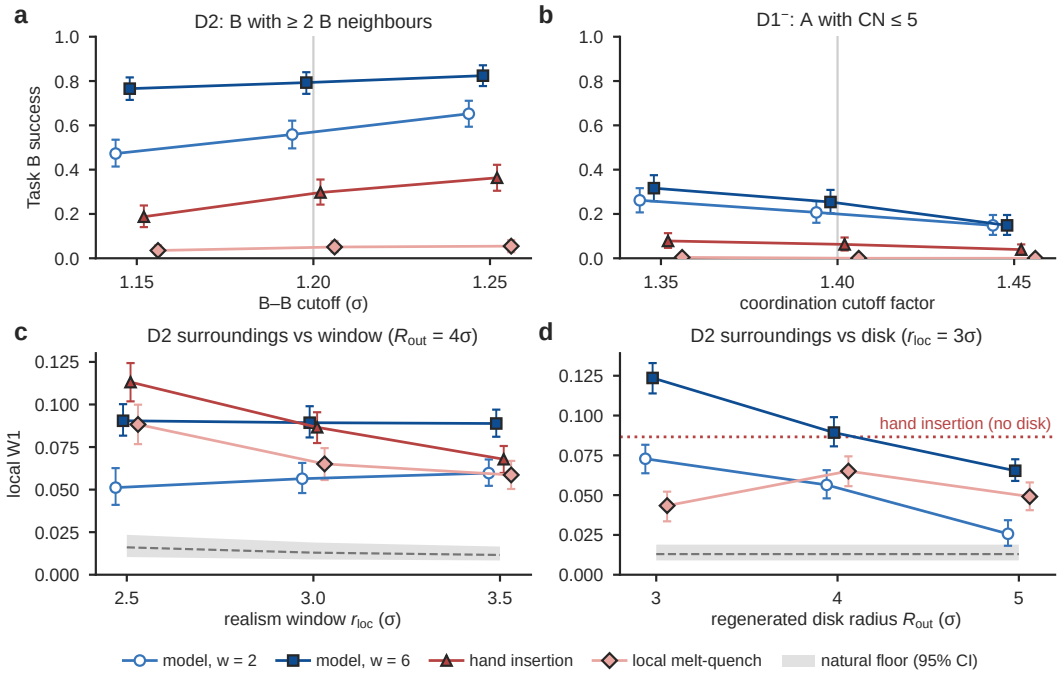


Figure 5: Sensitivity (seed 0, Task B). **(a, b)** Success against the defect cutoff for D2 and D1$^-$ (grey line: frozen value). **(c, d)** D2 local W1 against the realism window and the regenerated-disk radius; grey band: natural floor. The method ranking holds everywhere; hand insertion's damage is local (c), and a larger disk gives more natural surroundings (d).

Table 4: Training noise range (Task B, D2, seed 0). "Kept" = success after / before relaxation.

Recipe	w	placed (before relax)	success (after)	kept	local W1
$\sigma_{max} = 0.5$	2	0.67	0.35	52%	0.040
$\sigma_{max} = 0.5$	6	0.94	0.54	57%	0.064
$\sigma_{max} = 1.0$	2	0.86	0.56	65%	0.056
$\sigma_{max} = 1.0$	6	1.00	0.79	79%	0.089